College of Computer & Information Sciences

IS241–Database Management and Application

Study Planner Tracker

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Phase 1:
Introduction

Description

The proposed information system is developed for University students who need an organized and centralized way to manage their academic schedules. Most students rely on multiple platforms such as digital notes, calendars, or regular planners, which can lead to confusion and missed academic deadlines. This system is designed as a student planner tracker that provides a unified and easy to use platform for managing all academic related activities in one place.

Purpose

The purpose of the student planner tracker is to improve students academic experience, by organizing their schedules efficiently and reduce issues related to missing due dates, overlapping tasks, and poor time management. The system aims to provide students with a clear overview of their courses, assignments, exams, and study plan, allowing them to track academic progress more effectively.

Goals

- Provide a single, unified place for managing all academic schedules and study plans.
- Help students keep track of assignments, exams, and study tasks.
- Reduce overlapping and missed due dates caused by poor schedule organization.
- Improve students' time management and study planning skills.

- Allow students to monitor their study progress in an organized manner.

Information Systems Scope

The scope of the student planner tracker includes:

- Managing student profiles and enrolled courses
- Recording and tracking assignments, exams, and their due dates.
- Organizing study schedules and academic tasks
- Providing a clear timeline view of upcoming academic activities

The system does not fall within the scope of features related to grading, instructor evaluation, or direct integration with university registration systems. It only covers student-side academic planning and organization.

Business rules

Student and course

Type of relationship: one to many or 1-M

Rule of Business:

- A Student may sign up for more than one course.
- There is only one Student associated with each course

Explanation of cardinality:

- A Student may have one course or several
- Only a single Student can be linked to a course

Courses and courseEvent

Type of relationship: One to many 1-M

Rule of business:

- Multiple courseEvents(exams,assignments,and projects) may be included in a single course.
- Every courseEvent is specific to a single course.

Explanation of the cardinality:

- There can be zero or many events in a course
- Every event needs to be connected to a single course

Courses and Topic

Type of relationship: one to many 1-M

Rule of business:

- Several topics can be covered in a single course
- Every topic is exclusive to a single course

An explanation of the cardinality:

- There must be at least one topic in a course
- Without a course, a topic cannot exist.

Student and availableBlocks

Type of relationship: One to many 1-M

Rule of business:

- Multiple AvailableBlocks(free time slots) can be defined by one single Student
- Every AvailableBlock is associated with a single Student

An explanation of cardinality:

- A Student available block can range from zero to several
- Each block must be associated with one block

AvailableBlock and StudySession

Type of relationship: one to many 1-M

Rule of business:

- One availableBlock can make many studySessions
- Each studySession is scheduled within one AvailableBlock

Explanation of cardinality:

- There could be a block that may have zero or many studySession
- Each of the study sessions must belong to one block

Topics and StudySession

Relationship type: One to many 1-M

Rule of business:

- There can be more than one study session for a single topic
- There is just one topic linked to each study session

Explanation of cardinality:

- A topic may be studied many times
- Each study session focuses on one topic

Student and StudyPlanItem

Relationship type:One to many 1-M

BusinessRule:

- One Student can do and create multiple StudyPlanItem
- Each StudyPlanItem is associated to exactly one Student

Explanation of cardinality:

- A Student may have zero or many studyPlan items
- Each study plan item must associated with one and only one Student

Topic and StudyPlanItem

Relationship type: one to many 1-M

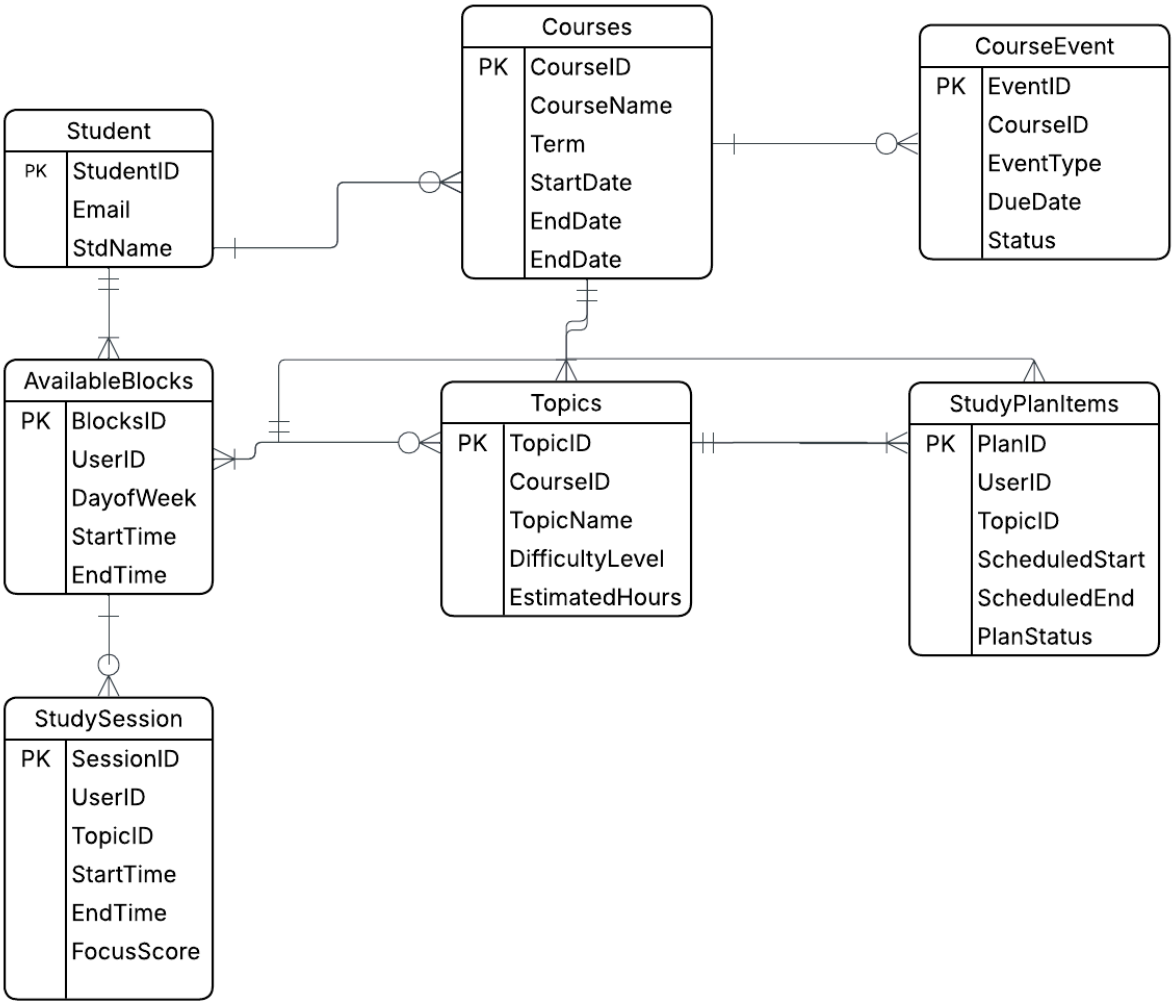
Business rule:

- One topic can be appeared in multiple StudyPlanItem
- Each StudyPlanItem refers to exactly one Topic

Explanation of cardinality:

- A topic may show in one or many studyPlanItem
- Each studyPlanItem must be associated with one and only one topic.

ERD Diagram


Courses
PK CourseID
CourseName
Term
StartDate
EndDate
EndDate
CourseEvent
PK EventID
CourseID
EventType
DueDate
Status
Student
PK StudentID
Email
StdName
AvailableBlocks
PK BlocksID
UserID
DayofWeek
StartTime
EndTime
Topics
PK TopicID
CourseID
TopicName
DifficultyLevel
EstimatedHours
StudyPlanItems
PK PlanID
UserID
TopicID
ScheduledStart
ScheduledEnd
PlanStatus
StudySession
PK SessionID
UserID
TopicID
StartTime
EndTime
FocusScore

Entity Descriptions & Data Requirements

Student

The Student entity represents each registered student who interacts with the system. It stores essential identification and authentication information required for personalization.

Data Requirements:

Primary Key (PK):

- Student_ID (int, NOT NULL, unique)

Attributes:

- STDName (varchar(100), NOT NULL)
- Email (varchar(150), NOT NULL, UNIQUE)

Constraints / Rules:

- Email must be unique for each user.

Courses

Represents a course entry created by a user with the Study Planner System.Stores the list of courses that exist in the system, each course record corresponds to a course that a specific user chooses to add for planning and tracking purpose.

Data Requirements:

PrimaryKey:

- CourseID (int, NOT NULL, unique)

Attributes:

- Student_ID (int, NOT NULL, unique)
- CourseName (varchar(120), NOT NULL)
- Term(int, NULL)
- StartDate(date, NOT NULL)
- EndDate(date, NOT NULL)

CoursesEvent

Represents a schedule of academic related-dates for a specific course to track and plan, such as an assignment, project or exam.

Data Requirements:

Primary Key:

- EventID(int, NOT NULL)

Foreign Keys :

- CourseID (int, NOT NULL)

Attributes:

- EventType(Varchar(20), NOT NULL)
- DueDate (date, NOT NULL)
- Status(varchar(20), NOT NULL)

Constraints / Rules:

- A Student cannot have overlapping events
- EventID is generated by default with a length of (5)

Topic

Represents the chapters or study topics associated within a course**,** including difficulty and estimated study hours.

Data Requirements:

Primary Key:

- TopicID (int, NOT NULL, unique)

Foreign Key:

- CourseID (int, NOT NULL)

Attributes:

- TopicName (varchar(150), NOT NULL)
- DifficultyLevel (varchar(1), NOT NULL)
- EstimatedHours (number(5,2), NOT NULL)

Constraints / Rules:

- EstimatedHours must be ≥ 0.
- Difficulty level must be between 1 and 5.
- A course cannot have duplicated topics with the same TopicName.

StudyPlanItems

Represents a scheduled study session created by a Student for a specific course.

Data Requirements:

Primary Key:

- PlanID (int, NOT NULL, unique)

Foreign Key:

- Student_ID (int, NOT NULL, unique)
- CourseID (int, NOT NULL)

Attributes:

- ScheduledStart (datetime, NOT NULL)
- ScheduledEnd (datetime, NOT NULL)
- PlanStatus (varchar(20), NOT NULL)

Constraints / Rules:

- ScheduledEnd must be after ScheduledStart.
- A Student can't have overlapping study sessions.

AvailableBlocks

Represents the weekly time blocks when the user is available to study.

Data Requirements:

Primary Key:

- BlocksID (int, NOT NULL, unique)

Foreign Key:

- Student_ID (int, NOT NULL, unique)

Attributes:

- DayofWeek (varchar(10), NOT NULL)
- StartTime (date, NOT NULL)
- EndTime (date, NOT NULL)

Constraints / Rules:

- EndTime must be after StartTime.

StudySession

Represents an actual study session completed by a user. It stores details about the course topic, the time duration, and Focus score.

Data Requirements:

Primary Key:

- SessionID (int, NOT NULL, unique)

Foreign Key:

- Student_ID (int, NOT NULL, unique)
- TopicID (int, NOT NULL)

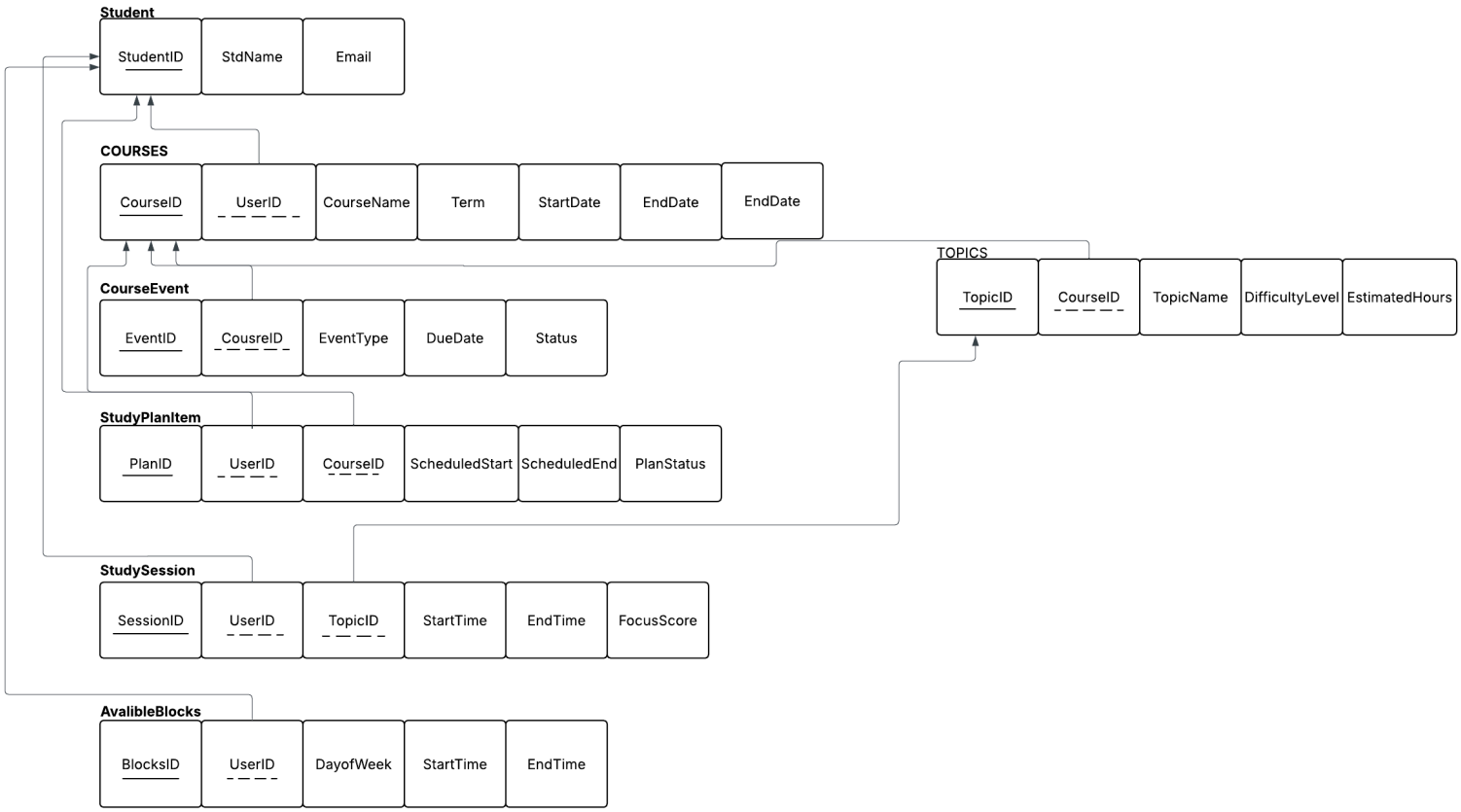
Attributes:

- StartTime (datetime, NOT NULL)
- EndTime (datetime, NOT NULL)
- FocusScore (varchar(1), NULL)

Constraints / Rules:

- EndTime must be after StartTime.
- FocusScore if provided should be within the allowed range (1-5).

Phase 2:
Mapping



Phase 3

SQL Script

Create Table:

```
create table course(
   course_ID  int not null,
   user_ID int not null,
   courseName varchar(120) not null,
   term int null,
   startDate date not null,
   endDate date not null,
primary key(course_ID),
foreign key (user_ID) references users (user_ID)
);

create table student(
Student_ID int not null ,
stdName varchar(100) not null,
email varchar(150) not null unique,
primary key (student_Id)
);

create table courseEvet(
   event_ID  number generated by default as identity,
   course_ID  int not null,
   eventType varchar2(20),
   dueDate date not null,
   status varchar2(20) not null,
   primary key (event_ID),
   foreign key (course_ID) references course(course_ID),
   check (eventType in ('Assignment','Project','MidTerm','Quiz','Final'))
   )
```

```
create table topic(
    plan_ID  int not null,
    course_ID  int not null,
    student_ID  int not null,
    ScheduledStart date NOT NULL,
    ScheduledEnd date NOT NULL,
    PlanStatus varchar2(20) NOT NULL,
    primary key (plan_ID),
    foreign key (course_ID) references course(course_ID),
     foreign key (student_ID) references student(student_ID),
    check (ScheduledEnd>ScheduledStart)
)

create table studyPlanItem(
    plan_ID  int not null,
    course_ID  int not null,
    student_ID  int not null,
    ScheduledStart date NOT NULL,
    ScheduledEnd date NOT NULL,
    PlanStatus varchar2(20) NOT NULL,
    primary key (plan_ID),
    foreign key (course_ID) references course(course_ID),
     foreign key (student_ID) references student(student_ID),
    check (ScheduledEnd>ScheduledStart))

    create table AvailableBlocks(
    blocks_ID  int not null,
    student_ID  int not null,
    DayofWeek varchar(10)  NOT NULL,
    StartTime date NOT NULL,
     EndTime date NOT NULL,
    primary key (blocks_ID),
     foreign key (student_ID) references student(student_ID),
    check (EndTime>StartTime)
)
```

```
create table StudySession(
   session_ID  int not null,
   student_ID  int not null,
   topic_ID  int not null,
   StartTime date NOT NULL,
   EndTime date NOT NULL,
focusScore varchar2(1),
   primary key (session_ID),
    foreign key (student_ID) references student(student_ID),
    foreign key (topic_ID) references topic(topic_ID),
   check (EndTime>StartTime),
   check (focusScore in ('1','2','3','4','5'))
)
)
```

Insert Data

// Data inserted in the student table

```
insert into student (STUDENT_ID,STDNAME,EMAIL)
values (2245, 'danah altuwaijri','dana@gmail.com');

insert into student (STUDENT_ID,STDNAME,EMAIL)
values (2234, 'shoug alshowaer','shoug@gmail.com');

insert into student (STUDENT_ID,STDNAME,EMAIL)
values (2235, 'Deema Alkasabi ','deema@gmail.com');

insert into student (STUDENT_ID,STDNAME,EMAIL)
values (2236, 'Yasmeen Alswailem ','yasmeen@gmail.com');

insert into student (STUDENT_ID,STDNAME,EMAIL)
values (2237, 'sara Alswailem ','sara@gmail.com');
```

// Data inserted in the course table
//student 2245 - danah altuwaijri - 2 courses

insert into course
(COURSE_ID,STUDENT_ID,COURSENAME,TERM,STARTDATE,ENDDATE)
values (210,2245, 'DataStructure and algorithm',2,
to_date('2026-01-1', 'YYYY-MM-DD'),to_date('2026-05-19', 'YYYY-MM-DD'));

insert into course
(COURSE_ID,STUDENT_ID,COURSENAME,TERM,STARTDATE,ENDDATE)
values (205,2245, 'Business Process Management',2,
to_date('2026-01-1', 'YYYY-MM-DD'),to_date('2026-05-19', 'YYYY-MM-DD'));

//student 2235 - deema alkasabi - 2 courses

insert into course
(COURSE_ID,STUDENT_ID,COURSENAME,TERM,STARTDATE,ENDDATE)
values (111,2235, 'Accounting',1,
to_date('2025-08-14', 'YYYY-MM-DD'),to_date('2025-12-20', 'YYYY-MM-DD'));

insert into course
(COURSE_ID,STUDENT_ID,COURSENAME,TERM,STARTDATE,ENDDATE)
values (103,2235, 'Islamic',2,
to_date('2026-01-01', 'YYYY-MM-DD'),to_date('2025-05-19', 'YYYY-MM-DD'));

//student 2236 - yasmeen alswailem - 2 courses

insert into course (course_id,student_id,coursename,term,startdate,enddate)
values (105,2236, 'Quran',2,
to_date('2026-01-01', 'YYYY-MM-DD'),to_date('2025-05-19', 'YYYY-MM-DD'));

insert into course (course_id,student_id,coursename,term,startdate,enddate)
values (101,2236, 'Arab',2,
to_date('2026-01-01', 'YYYY-MM-DD'),to_date('2025-05-19', 'YYYY-MM-DD'));

//student 2234 - shoug alshowaer - 2 courses

insert into course (course_id,student_id,coursename,term,startdate,enddate)

```
values (241,2234, 'Database Management and Applications',2,
to_date('2026-01-1', 'YYYY-MM-DD'),to_date('2026-05-19', 'YYYY-MM-DD'));

insert into course (course_id,student_id,coursename,term,startdate,enddate)
values (330,2234, 'Introduction to Operating Systems',2,
to_date('2026-01-1', 'YYYY-MM-DD'),to_date('2026-05-19', 'YYYY-MM-DD'));
```

//student 2237 - sara alswailem - 1 course

```
insert into course (course_id,student_id,coursename,term,startdate,enddate)
values (331,2237, 'English',2,
to_date('2026-01-1', 'YYYY-MM-DD'),to_date('2026-05-19', 'YYYY-MM-DD'));
```

//Student 2234 Data

```
insert into studyPlanItem
values (601, 241, 2234, to_date('2026-03-03 09:00','YYYY-MM-DD
HH24:MI'),to_date('2026-03-03 11:00','YYYY-MM-DD HH24:MI'),'Planned');

insert into studyPlanItem
values (602, 330, 2234,to_date('2026-03-04 10:00','YYYY-MM-DD HH24:MI'),
to_date('2026-03-04 12:00','YYYY-MM-DD HH24:MI'), 'Completed');

insert into AvailableBlocks
values (701, 2234, 'Sunday', to_date('2026-03-01 08:00','YYYY-MM-DD
HH24:MI'),to_date('2026-03-01 12:00','YYYY-MM-DD HH24:MI'));

insert into AvailableBlocks
values (702, 2234, 'Monday', to_date('2026-03-02 10:00','YYYY-MM-DD HH24:MI'),
to_date('2026-03-02 14:00','YYYY-MM-DD HH24:MI'));

insert into studysession (session_id,student_id,topic_id,starttime,endtime,     focuscore)
values ( 801, 2234, (select topic_id from topic where topicname = 'sql'), to_date('2026-03-01
10:00','yyyy-mm-dd hh24:mi'), to_date('2026-03-01 12:00','yyyy-mm-dd hh24:mi'), 5);
```

//Student 2235

```
insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (802, 2235, (select topic_id from topic where topicname = 'os'), to_date('2026-03-02
11:00','yyyy-mm-dd hh24:mi'), to_date('2026-03-02 13:00','yyyy-mm-dd hh24:mi'), 4);

insert into courseEvet (course_ID, eventType, dueDate, status)
values (111, 'Assignment', to_date('2025-10-10','YYYY-MM-DD'), 'Completed');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (111, 'MidTerm', to_date('2025-11-01','YYYY-MM-DD'), 'Completed');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (103, 'Project', to_date('2026-04-15','YYYY-MM-DD'), 'Pending');

insert into topic (course_id, topicname, difficultylevel, estimatedhours)
values (111, 'financial accounting', '3', 2.00);

insert into topic (course_id, topicname, difficultylevel, estimatedhours)
values (103, 'hadith', '4', 3.00);

insert into studyPlanItem
values (345, 111, 2235, to_date('2025-10-05 09:00','YYYY-MM-DD HH24:MI'),
to_date('2025-10-05 11:00','YYYY-MM-DD HH24:MI'),
'Completed');

insert into studyPlanItem
values (346, 103, 2235, to_date('2026-04-10 14:00','YYYY-MM-DD HH24:MI'),
to_date('2026-04-10 16:00','YYYY-MM-DD HH24:MI'), 'Planned');

insert into AvailableBlocks
values (901, 2235, 'Tuesday', to_date('2025-10-01 08:00','YYYY-MM-DD HH24:MI'),
to_date('2025-10-01 12:00','YYYY-MM-DD HH24:MI'));

insert into AvailableBlocks
values (902, 2235, 'Wednesday', to_date('2025-10-02 10:00','YYYY-MM-DD HH24:MI'),
to_date('2025-10-02 14:00','YYYY-MM-DD HH24:MI'));
```

```
insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (901, 2235, (select topic_id from topic where topicname = 'financial accounting'),
to_date('2025-10-01 09:00','yyyy-mm-dd hh24:mi'), to_date('2025-10-01 11:00','yyyy-mm-dd
hh24:mi'), 5);

insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (902, 2235, (select topic_id from topic where topicname = 'hadith'), to_date('2026-04-10
14:00','yyyy-mm-dd hh24:mi'), to_date('2026-04-10 16:00','yyyy-mm-dd hh24:mi'), 4);
```

//Student 2245

```
insert into courseEvet (course_ID, eventType, dueDate, status)
values (210, 'Assignment', to_date('2026-02-15','YYYY-MM-DD'), 'Pending');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (210, 'MidTerm', to_date('2026-03-15','YYYY-MM-DD'), 'Pending');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (205, 'Project', to_date('2026-04-20','YYYY-MM-DD'), 'Pending');

insert into topic (course_id, topicname, difficultylevel, estimatedhours)
values (210, 'recursion', '4', 3.00);

insert into topic (course_id, topicname, difficultylevel, estimatedhours)
values (205, 'bpmn', '3', 2.00);

insert into studyPlanItem
values (567, 210, 2245, to_date('2026-02-10 09:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-10 12:00','YYYY-MM-DD HH24:MI'),
'Planned');

insert into studyPlanItem
values (568, 205, 2245, to_date('2026-04-15 14:00','YYYY-MM-DD HH24:MI'),
to_date('2026-04-15 16:00','YYYY-MM-DD HH24:MI'),
'Planned');

insert into AvailableBlocks
values (1001, 2245, 'Wednesday', to_date('2026-02-11 08:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-11 12:00','YYYY-MM-DD HH24:MI'));
```

```
insert into AvailableBlocks
values (1002, 2245, 'Thursday', to_date('2026-02-12 10:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-12 14:00','YYYY-MM-DD HH24:MI'));

insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (1001, 2245,
  (select topic_id from topic where topicname = 'recursion'),
  to_date('2026-02-11 09:00','yyyy-mm-dd hh24:mi'),
  to_date('2026-02-11 12:00','yyyy-mm-dd hh24:mi'),
  5);

insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (1002, 2245,
  (select topic_id from topic where topicname = 'bpmn'),
  to_date('2026-04-15 14:00','yyyy-mm-dd hh24:mi'),
  to_date('2026-04-15 16:00','yyyy-mm-dd hh24:mi'),
  4);
```

//Student 2236

```
insert into courseEvet (course_ID, eventType, dueDate, status)
values (105, 'Assignment', to_date('2026-02-20','YYYY-MM-DD'), 'Pending');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (105, 'MidTerm', to_date('2026-03-20','YYYY-MM-DD'), 'Pending');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (101, 'Project', to_date('2026-04-25','YYYY-MM-DD'), 'Pending');

insert into topic (course_id, topicname, difficultylevel, estimatedhours)
values (105, 'tajweed', '4', 3.00);

insert into topic (course_id, topicname, difficultylevel, estimatedhours)
values (101, 'arabic grammar', '3', 2.00);

insert into studyPlanItem
values (789, 105, 2236, to_date('2026-02-15 09:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-15 12:00','YYYY-MM-DD HH24:MI'),
```

'Planned');

insert into studyPlanItem
values (790, 101, 2236, to_date('2026-04-20 14:00','YYYY-MM-DD HH24:MI'),
to_date('2026-04-20 16:00','YYYY-MM-DD HH24:MI'),
'Planned');

insert into AvailableBlocks
values (1101, 2236, 'Tuesday', to_date('2026-02-17 08:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-17 12:00','YYYY-MM-DD HH24:MI'));

insert into AvailableBlocks
values (1102, 2236, 'Friday', to_date('2026-02-18 10:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-18 14:00','YYYY-MM-DD HH24:MI'));

insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (1101, 2236, (select topic_id from topic where topicname = 'tajweed'),
to_date('2026-02-17 09:00','yyyy-mm-dd hh 24:mi'), to_date('2026-02-17 12:00','yyyy-mm-dd hh24:mi'), 5);

insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (1102, 2236, (select topic_id from topic where topicname = 'arabic grammar'),
  to_date('2026-02-18 10:00','yyyy-mm-dd hh24:mi'), to_date('2026-02-18 14:00','yyyy-mm-dd hh24:mi'), 4);

//Student 2237

insert into courseEvet (course_ID, eventType, dueDate, status)
values (331, 'Assignment', to_date('2026-02-25','YYYY-MM-DD'), 'Pending');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (331, 'MidTerm', to_date('2026-03-25','YYYY-MM-DD'), 'Pending');

insert into courseEvet (course_ID, eventType, dueDate, status)
values (331, 'Project', to_date('2026-04-30','YYYY-MM-DD'), 'Pending');

insert into topic (course_id, topicname, difficultylevel, estimatedhours)
values (331, 'english literature', '3', 2.00);

```
insert into studyPlanItem
values (890, 331, 2237, to_date('2026-02-20 09:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-20 12:00','YYYY-MM-DD HH24:MI'),
'Planned');

insert into AvailableBlocks
values (1201, 2237, 'Monday', to_date('2026-02-23 08:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-23 12:00','YYYY-MM-DD HH24:MI'));

insert into AvailableBlocks
values (1202, 2237, 'Thursday', to_date('2026-02-24 10:00','YYYY-MM-DD HH24:MI'),
to_date('2026-02-24 14:00','YYYY-MM-DD HH24:MI'));

insert into studysession (session_id,student_id,topic_id,starttime,endtime,focusscore)
values (1201, 2237,  (select topic_id from topic where topicname = 'english literature'),
to_date('2026-02-23 09:00','yyyy-mm-dd hh24:mi'), to_date('2026-02-23 12:00','yyyy-mm-dd
hh24:mi'), 5);
```

UPDATE:

```
update course set COURSENAME = 'Operating Systems' where COURSE_ID = 330;

update student set email = 'shoug.alshowaer@example.com' where STUDENT_ID = 2234;

update courseEvet set eventType = 'MidTerm' where event_ID = 1;

update topic set PlanStatus = 'Completed' where plan_ID = 1;

update studyPlanItem set PlanStatus = 'Completed' where plan_ID = 1;
```

DELETE:

```
delete studysession where session_ID = 1;

delete AvailableBlocks where blocks_ID = 701;
```

```
delete studyPlanItem where plan_ID = 602;

delete courseEvet where event_ID = 1;
```

JOIN:

```
select s.stdName, c.courseName
from student s
join course c on s.student_ID = c.student_ID;

select s.stdName, c.courseName, ce.eventType, ce.dueDate
from student s
join course c on s.student_ID = c.student_ID
join courseEvet ce on c.course_ID = ce.course_ID
where ce.status = 'Pending';

select s.stdName, c.courseName, t.topicName, t.difficultyLevel
from student s
join course c on s.student_ID = c.student_ID
join topic t on c.course_ID = t.course_ID
where t.difficultyLevel >= 3;
```

Group By:

```
select student_ID, count(*) as total_courses
from course
group by student_ID;
```

```
select student_ ID
from course
group by student_ID having count (*) < 2;
```

DROP:

```
Drop table course casecade constraints ;
Drop table student  casecade constraints ;
Drop table courseEvent casecade constraints ;
Drop table studyPlanItem casecade constraints ;
Drop table studeSession casecade constraints ;
Drop table topic casecade constraints ;
Drop table availableBlocks casecade constraints ;
```

Database Table

STUDENT

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
STUDENT_ID	NUMBER	N		1		
STDNAME	VARCHAR2(100 BYTE)	N				
EMAIL	VARCHAR2(150 BYTE)	N				

COURSE

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
COURSE_ID	NUMBER	N		1		
STUDENT_ID	NUMBER	N				
COURSENAME	VARCHAR2(120 BYTE)	N				
TERM	NUMBER	Y				
STARTDATE	DATE	N				
ENDDATE	DATE	N				

COURSEEVET

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
EVENT_ID	NUMBER	N	"WKSP_IS241DANAHALTUWAIJRI"."ISE...	1		BY DEFAULT
COURSE_ID	NUMBER	N				
EVENTTYPE	VARCHAR2(20 BYTE)	Y				
DUEDATE	DATE	N				
STATUS	VARCHAR2(20 BYTE)	N				

AVAILABLEBLOCKS

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
BLOCKS_ID	NUMBER	N		1		
STUDENT_ID	NUMBER	N				
DAYOFWEEK	VARCHAR2(10 BYTE)	N				
STARTTIME	DATE	N				
ENDTIME	DATE	N				

STUDYPLANITEM

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
PLAN_ID	NUMBER	N		1		
COURSE_ID	NUMBER	N				
STUDENT_ID	NUMBER	N				
SCHEDULEDSTART	DATE	N				
SCHEDULEDEND	DATE	N				
PLANSTATUS	VARCHAR2(20 BYTE)	N				

STUDYSESSION

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
SESSION_ID	NUMBER	N		1		
STUDENT_ID	NUMBER	N				
TOPIC_ID	NUMBER	N				
STARTTIME	DATE	N				
ENDTIME	DATE	N				
FOCUSSCORE	VARCHAR2(1 BYTE)	Y				

TOPIC

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
TOPIC_ID	NUMBER	N	"WKSP_IS241DANAHALTUWAIJRI"."ISE...	1		BY DEFAULT
COURSE_ID	NUMBER	N				
TOPICNAME	VARCHAR2(120 BYTE)	N				
DIFFICULTYLEVEL	VARCHAR2(1 BYTE)	Y				
ESTIMATEDHOURS	NUMBER(5,2)	N				

Inserted Data

COURSE

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

COURSE_ID	STUDENT_ID	COURSENAME	TERM	STARTDATE	ENDDATE
241	2234	Database Management	2	1/1/2026	5/19/2026
330	2234	Operating Systems	2	1/1/2026	5/19/2026
111	2235	Accounting	1	8/14/2025	12/20/2025
105	2236	Quran	2	1/1/2026	5/19/2025
205	2245	Business Process Management	2	1/1/2026	5/19/2026
103	2235	Islamic	2	1/1/2026	5/19/2025
331	2237	English	2	1/1/2026	5/19/2026
210	2245	DataStructure and algorthim	2	1/1/2026	5/19/2026
101	2236	Arab	2	1/1/2026	5/19/2025

COURSEEVET

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

EVENT_ID	COURSE_ID	EVENTTYPE	DUEDATE	STATUS
2	241	MidTerm	4/1/2026	Completed
23	330	Project	4/20/2026	Pending
44	210	Assignment	2/15/2026	Pending
45	210	MidTerm	3/15/2026	Pending
46	205	Project	4/20/2026	Pending
47	105	Assignment	2/20/2026	Pending
48	105	MidTerm	3/20/2026	Pending
49	101	Project	4/25/2026	Pending
50	331	Assignment	2/25/2026	Pending
51	331	MidTerm	3/25/2026	Pending
52	331	Project	4/30/2026	Pending
41	111	Assignment	10/10/2025	Completed
42	111	MidTerm	11/1/2025	Completed
43	103	Project	4/15/2026	Pending

STUDENT

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

STUDENT_ID	STDNAME	EMAIL
2245	danah altuwaijri	dana@gmail.com
2237	sara Alswailem	sara@gmail.com
2235	Deema Alkasabi	deema@gmail.com
2236	Yasmeen Alswailem	yasmeen@gmail.com
2234	shoug alshowaer	shoug.alshowaer@example.com

TOPIC

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

TOPIC_ID	COURSE_ID	TOPICNAME	DIFFICULTYLEVEL	ESTIMATEDHOURS
1	241	SQL	3	2
2	330	OS	4	3
23	210	Recursion	4	3
24	205	BPMN	3	2
25	105	Tajweed	4	3
26	101	Arabic Grammar	3	2
27	331	English Literature	3	2
21	111	Financial Accounting	3	2
22	103	Hadith	4	3

STUDYPLANITEM

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Insert Row | Columns... | Filter... | Count Rows | Load Data | Download | Refresh

PLAN_ID	COURSE_ID	STUDENT_ID	SCHEDULEDSTART	SCHEDULEDEND	PLANSTATUS
567	210	2245	2/10/2026	2/10/2026	Planned
568	205	2245	4/15/2026	4/15/2026	Planned
789	105	2236	2/15/2026	2/15/2026	Planned
790	101	2236	4/20/2026	4/20/2026	Planned
890	331	2237	2/20/2026	2/20/2026	Planned
345	111	2235	10/5/2025	10/5/2025	Completed
346	103	2235	4/10/2026	4/10/2026	Planned

AVAILABLEBLOCKS

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
BLOCKS_ID	NUMBER	N		1		
STUDENT_ID	NUMBER	N				
DAYOFWEEK	VARCHAR2(10 BYTE)	N				
STARTTIME	DATE	N				
ENDTIME	DATE	N				

STUDYSESSION

Columns | Data | Indexes | Constraints | Grants | Statistics | Triggers | Dependencies | DDL | Sample Queries

+ Add Column | Modify Column | Rename Column | Drop Column | UI Defaults | Refresh | More

Column Name	Data Type	Nullable	Default	Primary Key	Comment	Identity
SESSION_ID	NUMBER	N		1		
STUDENT_ID	NUMBER	N				
TOPIC_ID	NUMBER	N				
STARTTIME	DATE	N				
ENDTIME	DATE	N				
FOCUSSCORE	VARCHAR2(1 BYTE)	Y				

Phase 4

SQL QUERIES

1. Display the Course_ID and DueDate for CourseEvent where the event type is ‘Project’

```
select course_ID, dueDate from courseEvet
where eventType = 'Project';
```

Results | Explain | Describe | Saved SQL | History

COURSE_ID	DUEDATE
330	4/20/2026
330	4/20/2026
205	4/20/2026
101	4/25/2026
331	4/30/2026
103	4/15/2026

2. Display the Course_ID, EventType, and DueDate for events due before April 1, 2026

```
select course_ID, eventType, dueDate from courseEvet
where dueDate < to_date('2026-04-01','YYYY-MM-DD');
```

Results | Explain | Describe | Saved SQL | History

COURSE_ID	EVENTTYPE	DUEDATE
241	Assignment	3/10/2026
210	Assignment	2/15/2026
210	MidTerm	3/15/2026
105	Assignment	2/20/2026
105	MidTerm	3/20/2026
331	Assignment	2/25/2026
331	MidTerm	3/25/2026
111	Assignment	10/10/2025
111	MidTerm	11/1/2025

3. Display the Course_ID, EventType, and DueDate for events due between March 1, 2026 and April 15, 2026

```
select course_ID, eventType, dueDate from courseEvet
where dueDate between to_date('2026-03-01','YYYY-MM-DD') and to_date('2026-04-30','YYYY-MM-DD');
```

Results | Explain | Describe | Saved SQL | History

COURSE_ID	EVENTTYPE	DUEDATE
241	Assignment	3/10/2026
241	MidTerm	4/1/2026
330	Project	4/20/2026
210	MidTerm	3/15/2026
205	Project	4/20/2026
105	MidTerm	3/20/2026
101	Project	4/25/2026
331	MidTerm	3/25/2026
331	Project	4/30/2026
103	Project	4/15/2026

10 rows returned in 0.02 seconds Download

4. Display each student name with their course name by connecting the student table with the course table using the Student_ID

```
select s.stdName, c.courseName
from student s
join course c on s.student_ID = c.student_ID;
```

Results | Explain | Describe | Saved SQL | History

STDNAME	COURSENAME
shoug alshowaer	Database Management and Applications
shoug alshowaer	Introduction to Operating Systems
Deema Alkasabi	Accounting
Yasmeen Alswailem	Quran
danah altuwaijri	Business Process Management
Deema Alkasabi	Islamic
sara Alswailem	English
danah altuwaijri	DataStructure and algorthim
Yasmeen Alswailem	Arab

9 rows returned in 0.01 seconds Download

5. Display the StdName, CourseName, EventType, and DueDate for CourseEvent that are ‘Pending’

```sql
select s.stdName, c.courseName, ce.eventType, ce.dueDate
from student s
join course c on s.student_ID = c.student_ID
join courseEvet ce on c.course_ID = ce.course_ID
where ce.status = 'Pending';
```

Results | Explain | Describe | Saved SQL | History

STDNAME	COURSENAME	EVENTTYPE	DUEDATE
shoug alshowaer	Database Management and Applications	Assignment	3/10/2026
shoug alshowaer	Introduction to Operating Systems	Project	4/20/2026
danah altuwaijri	DataStructure and algorthim	Assignment	2/15/2026
danah altuwaijri	DataStructure and algorthim	MidTerm	3/15/2026
danah altuwaijri	Business Process Management	Project	4/20/2026
Yasmeen Alswailem	Quran	Assignment	2/20/2026
Yasmeen Alswailem	Quran	MidTerm	3/20/2026
Yasmeen Alswailem	Arab	Project	4/25/2026
sara Alswailem	English	Assignment	2/25/2026
sara Alswailem	English	MidTerm	3/25/2026

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.02 seconds Download

6. Display the CourseName and Difficulty level for topics that are difficult (>=3), for each student’s courses.

```sql
select s.stdName, c.courseName, t.topicName, t.difficultyLevel
from student s
join course c on s.student_ID = c.student_ID
join topic t on c.course_ID = t.course_ID
where t.difficultyLevel >= 3;
```

Results | Explain | Describe | Saved SQL | History

STDNAME	COURSENAME	TOPICNAME	DIFFICULTYLEVEL
shoug alshowaer	Database Management and Applications	SQL	3
shoug alshowaer	Introduction to Operating Systems	OS	4
danah altuwaijri	DataStructure and algorthim	Recursion	4
danah altuwaijri	Business Process Management	BPMN	3
Yasmeen Alswailem	Quran	Tajweed	4
Yasmeen Alswailem	Arab	Arabic Grammar	3
sara Alswailem	English	English Literature	3
Deema Alkasabi	Accounting	Financial Accounting	3
Deema Alkasabi	Islamic	Hadith	4

9 rows returned in 0.02 seconds Download

7. Display how many courses each student has.

```
select student_ID, count(*) as total_courses from course group by student_ID;
```

Results | Explain | Describe | Saved SQL | History

STUDENT_ID	TOTAL_COURSES
2234	2
2235	2
2236	2
2245	2
2237	1

5 rows returned in 0.01 seconds Download

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8. Display students who have less than 2 courses.

```
select student_ID
from course
group by student_ID having count (*) < 2;
```

Results | Explain | Describe | Saved SQL | History

STUDENT_ID
2237

1 rows returned in 0.00 seconds Download

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